MAPS4U STREET MAP SYSTEM (maps4u-maps) - COMPREHENSIVE TESTING REC

Release: v1.0.3 | Gates G1-G6 + v1.0.1-v1.0.3: ALL PASSED | Tests: 97/97 Passed | Coverage: 91% (3,238 stmts)

1. EXECUTIVE SMOKE, HARDENING & VERIFICATION SUMMARY

Suite / Check	Command	Result	Key Verified Metrics
End-to-End Smoke Test	python -m scripts.smoke_test	PASS	status=proof_ready, cache_hit=true,
Byte & Content Determinism	python -m scripts.verify_determinism	PASS	deterministic=true, 100% identical
Hot SQLite + Bundle Drill	python -m scripts.backup_restore_drill	PASS	restored_ok=true, manifest_sha256=t
Live Overpass 3-Area Pilots	python scripts/run_live_pilots.py	PASS	3/3 live QLD coastal proofs finaliz
Static Linter & Type Check	ruff check + mypy + pip check	PASS	0 lint errors, 0 type issues, 0 bro
Full Offline Pytest + Coverage	pytest -m 'not live' -v --cov	PASS	97 passed, 0 failed in 46.35s; 91%

2. STAGE GATE & RELEASE PROGRESSION (G1 -> v1.0.3)

Gate/Ver	Tag	Scope & Deliverables Verified	Tests
G1-G4	v0.1.0..v0.4.0	FastAPI/Celery, SQLite WAL, 6 templates, CairoCanvas PDF 1.4, STRtree labels	54 passed
G5-G6	v1.0.0-mvp	11-stage job executor, review/approve lock, reproduce(), Dockerfile + compose	61 passed
v1.0.1	v1.0.1	BUG-001..015 remediation: wheel resources, RBAC, project revision scoping	83 passed
v1.0.2	v1.0.2	SEC-001..003, DATA-001..003, QUEUE-001..003, QA-001, OPS-001 hardening	94 passed
v1.0.3	v1.0.3	Same-name spacing (52mm), commercial POI filter, legend swatches, Street Index	97 passed

3. LIVE OPENSTREETMAP OVERPASS THREE-PILOT ACCEPTANCE (v1.0.3)

Pilot Area	Format	Scale	Live Payload	Blockers	Overlaps	reproduce()	Locked pro
Maroochydore (MCY Reference)	A4-landscape	1:12,500	1,043,737 B	0	0 / 0	MATCHED	654171e199
Noosa Heads	A3-landscape	1:7,500	386,763 B	0	0 / 0	MATCHED	cf80b43d0d
Bribie Island - Sandstone Point	A4-portrait	1:10,000	368,274 B	0	0 / 0	MATCHED	6ecd72c9ab

4. SECURITY, PROVENANCE & QUEUE HARDENING MATRIX (v1.0.1 - v1.0.3)

- SEC-001..003: Fail-closed prod token policy (>=16 chars), distinct operator/reviewer tokens, header-only auth, immutable r
- DATA-001..003: Content-addressed OSM snapshots (SHA-256(bbox+query+raw_sha256)), frozen_inputs/ archive, GeoPackage corrup
- QUEUE-001..003 & OPS-001: Atomic claim_job_for_execution, exponential retry backoff, transient vs terminal Celery errors,
- Cartographic Polish (v1.0.3): min_same_name_distance_mm=52.0, commercial motel filtering, visual legend swatches, rendered

```
tests/api/test_api_phase0.py::test_system_health_ready_and_meta PASSED   [  1%]
tests/api/test_api_phase0.py::test_projects_and_geofence_revisions_lifecycle PASSED [  2%]
tests/api/test_api_phase0.py::test_illegal_geofence_rejected_with_error_code PASSED [  3%]
tests/api/test_stage2_viability_and_ui.py::test_post_viability_returns_assessment_for_valid_geofence PASSED [  4
tests/api/test_stage2_viability_and_ui.py::test_post_viability_blocks_when_min_font_override_exceeds_style_font
tests/api/test_stage2_viability_and_ui.py::test_post_viability_rejects_self_intersecting_geometry_with_422 PASSE
tests/api/test_stage2_viability_and_ui.py::test_geofence_revisions_list_and_static_ui_served PASSED [  7%]
tests/api/test_stage5_jobs_and_review.py::test_s5_t1_end_to_end_job_run_artifacts_approve_reject_and_corrections
tests/integration/test_acquire.py::test_build_overpass_query_normalises_bbox PASSED [  9%]
tests/integration/test_acquire.py::test_full_acquire_all_layers_and_reload PASSED [ 10%]
tests/integration/test_acquire.py::test_overpass_retry_ladder_on_failure PASSED [ 11%]
tests/integration/test_stage3_proof_render.py::test_s3_t3_end_to_end_vector_proof_render_and_structural_checks P
tests/integration/test_stage4_qa_and_bundle.py::test_s4_t3_qa_preflight_and_governed_manifest_bundle PASSED [ 13
tests/integration/test_stage5_reproducibility_and_pilots.py::test_s5_t2_and_t3_pilot_runs_and_reproducibility[Ma
tests/integration/test_stage5_reproducibility_and_pilots.py::test_s5_t2_and_t3_pilot_runs_and_reproducibility[No
tests/integration/test_stage5_reproducibility_and_pilots.py::test_s5_t2_and_t3_pilot_runs_and_reproducibility[Br
tests/integration/test_stage6_packaging_and_handover.py::test_s6_t1_docker_and_compose_packaging PASSED [ 17%]
tests/integration/test_stage6_packaging_and_handover.py::test_s6_t2_handover_docs_and_backup_restore_drill PASSE
tests/integration/test_stage6_packaging_and_handover.py::test_s6_t3_full_uat_operator_journey PASSED [ 19%]
tests/integration/test_v101_defect_remediation.py::test_bug001_runtime_dependencies_declared_and_pdf_render_work
tests/integration/test_v101_defect_remediation.py::test_bug002_resource_paths_installation_safe_outside_repo PAS
tests/integration/test_v101_defect_remediation.py::test_bug003_authentication_and_rbac_enforced PASSED [ 22%]
tests/integration/test_v101_defect_remediation.py::test_bug004_celery_queue_mode_dispatches_and_worker_executes
tests/integration/test_v101_defect_remediation.py::test_bug005_redis_env_var_standardised PASSED [ 24%]
tests/integration/test_v101_defect_remediation.py::test_bug006_duplicate_submission_rejected_and_reproduce_endpo
tests/integration/test_v101_defect_remediation.py::test_bug007_geofence_revision_ids_scoped_per_project PASSED [
tests/integration/test_v101_defect_remediation.py::test_bug008_failed_qa_and_corrupted_pdf_block_proof_ready_and
tests/integration/test_v101_defect_remediation.py::test_bug009_unexpected_pipeline_exceptions_transition_to_fail
tests/integration/test_v101_defect_remediation.py::test_bug009_unexpected_pipeline_exceptions_transition_to_fail
tests/integration/test_v101_defect_remediation.py::test_bug009_unexpected_pipeline_exceptions_transition_to_fail
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tests/integration/test_v101_defect_remediation.py::test_bug009_unexpected_pipeline_exceptions_transition_to_fail
tests/integration/test_v101_defect_remediation.py::test_bug009_unexpected_pipeline_exceptions_transition_to_fail
tests/integration/test_v101_defect_remediation.py::test_bug011_and_bug012_makefile_and_settings_field_names PASS
tests/integration/test_v101_defect_remediation.py::test_bug013_style_selection_validation_and_custom_style_manif
tests/integration/test_v101_defect_remediation.py::test_bug014_browser_ui_complete_operator_journey PASSED [ 41%
tests/integration/test_v101_defect_remediation.py::test_bug015_overpass_concurrency_cache_ttl_and_artifact_tampe
tests/integration/test_v102_production_hardening.py::test_sec001_production_rejects_missing_weak_or_known_fallba
tests/integration/test_v102_production_hardening.py::test_sec002_verified_reviewer_identities_cannot_be_forged P
tests/integration/test_v102_production_hardening.py::test_sec003_url_query_string_authentication_removed PASSED
tests/integration/test_v102_production_hardening.py::test_data001_immutable_osm_provenance_across_cache_refreshe
tests/integration/test_v102_production_hardening.py::test_data002_frozen_reproducibility_inputs_detect_or_isolat
tests/integration/test_v102_production_hardening.py::test_data003_geopackage_corruption_recovery_and_diagnostic_
tests/integration/test_v102_production_hardening.py::test_queue001_duplicate_dispatch_and_atomic_job_claiming PA
tests/integration/test_v102_production_hardening.py::test_queue002_retry_executes_or_enqueues_and_tracks_retry_c
tests/integration/test_v102_production_hardening.py::test_queue003_celery_task_outcomes_distinguish_transient_an
tests/integration/test_v102_production_hardening.py::test_qa001_no_runtime_qa_shortcuts_in_state_transitions PAS
tests/integration/test_v102_production_hardening.py::test_ops001_meaningful_readiness_checks_db_broker_and_worke
tests/integration/test_v102_production_hardening.py::test_cache_purge_protects_referenced_and_active_snapshots P
tests/unit/test_corrections.py::test_apply_valid_corrections_rename_reclass_add_remove PASSED [ 55%]
tests/unit/test_corrections.py::test_corrections_unknown_osm_id_and_stale_target_warnings PASSED [ 56%]
tests/unit/test_corrections.py::test_corrections_invalid_yaml_or_schema_raises PASSED [ 57%]
tests/unit/test_fetch_roads_bbox.py::test_fetch_roads_inside_bounding_box_and_cache PASSED [ 58%]
tests/unit/test_stage3_canvas_and_style.py::test_s3_t1_cairo_canvas_deterministic_vector_pdf_and_thumbnail PASSE
tests/unit/test_stage3_canvas_and_style.py::test_s3_t2_load_style_yaml_and_reject_invalid PASSED [ 60%]
tests/unit/test_stage3_canvas_and_style.py::test_s3_t2_scale_band_selection_and_z_ordered_draw_plan PASSED [ 61%
tests/unit/test_stage4_labels_and_furniture.py::test_s4_t1_label_placement_determinism_collision_and_repeats PAS
tests/unit/test_stage4_labels_and_furniture.py::test_s4_t2_and_t4_compose_furniture_all_templates_and_insets[A3-
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tests/unit/test_stage4_labels_and_furniture.py::test_s4_t2_and_t4_compose_furniture_all_templates_and_insets[A3-
tests/unit/test_stage4_labels_and_furniture.py::test_s4_t2_and_t4_compose_furniture_all_templates_and_insets[A4-
tests/unit/test_stage4_labels_and_furniture.py::test_s4_t2_and_t4_compose_furniture_all_templates_and_insets[A4-
tests/unit/test_stage4_labels_and_furniture.py::test_s4_t2_and_t4_compose_furniture_all_templates_and_insets[A5-
tests/unit/test_stage4_labels_and_furniture.py::test_s4_t2_and_t4_compose_furniture_all_templates_and_insets[A5-
tests/unit/test_stage4_labels_and_furniture.py::test_nan_names_are_filtered_from_labels PASSED [ 70%]
tests/unit/test_stage4_labels_and_furniture.py::test_v103_same_name_distance_deduplication_and_commercial_poi_fi
tests/unit/test_state_and_db.py::test_migrations_are_idempotent PASSED   [ 72%]
tests/unit/test_state_and_db.py::test_state_machine_transitions_and_governed_outputs PASSED [ 73%]
tests/unit/test_state_and_db.py::test_inline_pipeline_runner_happy_path PASSED [ 74%]
tests/unit/test_state_and_db.py::test_recording_canvas_deterministic_drawlog PASSED [ 75%]
tests/unit/test_templates_and_assess.py::test_all_six_paper_templates_resolve_valid_map_area PASSED [ 76%]
tests/unit/test_templates_and_assess.py::test_assess_known_extent_produces_expected_scale_denominator PASSED [ 7
tests/unit/test_templates_and_assess.py::test_assess_undersized_font_config_is_blocked_with_remediation_choices
tests/unit/test_templates_and_assess.py::test_assess_dense_synthetic_fixture_triggers_density_warning PASSED [ 7
tests/unit/test_templates_and_assess.py::test_all_pilot_geofences_pass_across_all_paper_orientations[A3-landscap
tests/unit/test_templates_and_assess.py::test_all_pilot_geofences_pass_across_all_paper_orientations[A3-landscap
tests/unit/test_templates_and_assess.py::test_all_pilot_geofences_pass_across_all_paper_orientations[A3-landscap
tests/unit/test_templates_and_assess.py::test_all_pilot_geofences_pass_across_all_paper_orientations[A3-portrait
tests/unit/test_templates_and_assess.py::test_all_pilot_geofences_pass_across_all_paper_orientations[A3-portrait
tests/unit/test_templates_and_assess.py::test_all_pilot_geofences_pass_across_all_paper_orientations[A3-portrait
tests/unit/test_templates_and_assess.py::test_all_pilot_geofences_pass_across_all_paper_orientations[A4-landscap
tests/unit/test_templates_and_assess.py::test_all_pilot_geofences_pass_across_all_paper_orientations[A4-landscap
tests/unit/test_templates_and_assess.py::test_all_pilot_geofences_pass_across_all_paper_orientations[A4-landscap
tests/unit/test_templates_and_assess.py::test_all_pilot_geofences_pass_across_all_paper_orientations[A4-portrait
tests/unit/test_templates_and_assess.py::test_all_pilot_geofences_pass_across_all_paper_orientations[A4-portrait
tests/unit/test_templates_and_assess.py::test_all_pilot_geofences_pass_across_all_paper_orientations[A4-portrait
tests/unit/test_templates_and_assess.py::test_all_pilot_geofences_pass_across_all_paper_orientations[A5-landscap
tests/unit/test_templates_and_assess.py::test_all_pilot_geofences_pass_across_all_paper_orientations[A5-landscap
tests/unit/test_templates_and_assess.py::test_all_pilot_geofences_pass_across_all_paper_orientations[A5-landscap
tests/unit/test_templates_and_assess.py::test_all_pilot_geofences_pass_across_all_paper_orientations[A5-portrait
tests/unit/test_templates_and_assess.py::test_all_pilot_geofences_pass_across_all_paper_orientations[A5-portrait
tests/unit/test_templates_and_assess.py::test_all_pilot_geofences_pass_across_all_paper_orientations[A5-portrait
tests/unit/test_templates_and_assess.py::test_scale_monotonicity_with_geofence_area PASSED [ 98%]
tests/unit/test_templates_and_assess.py::test_hypothesis_scale_monotonicity_with_area PASSED [100%]
  /a0/usr/workdir/maps4u-street-map-system/src/maps4u/render/style_engine.py:241: DeprecationWarning: The 'shape
  /a0/usr/workdir/maps4u-street-map-system/src/maps4u/labels/engine.py:132: DeprecationWarning: The 'shapely.ops
  /a0/usr/workdir/maps4u-street-map-system/src/maps4u/labels/engine.py:195: DeprecationWarning: The 'shapely.ops
  /a0/usr/workdir/maps4u-street-map-system/src/maps4u/labels/engine.py:216: DeprecationWarning: The 'shapely.ops
  /a0/usr/workdir/maps4u-street-map-system/src/maps4u/labels/engine.py:240: DeprecationWarning: The 'shapely.ops
  /a0/usr/workdir/maps4u-street-map-system/src/maps4u/labels/engine.py:265: DeprecationWarning: The 'shapely.ops
tests/integration/test_stage6_packaging_and_handover.py::test_s6_t3_full_uat_operator_journey
tests/integration/test_stage6_packaging_and_handover.py::test_s6_t3_full_uat_operator_journey
tests/integration/test_stage6_packaging_and_handover.py::test_s6_t3_full_uat_operator_journey
Name                                    Stmts   Miss  Cover
-----------------------------------------------------------
src/maps4u/__init__.py                      1      0   100%
src/maps4u/api/__init__.py                  0      0   100%
src/maps4u/api/deps.py                    118     19    84%
src/maps4u/api/main.py                     48      1    98%
src/maps4u/api/routes_jobs.py             220     27    88%
src/maps4u/api/routes_projects.py          69      6    91%
src/maps4u/api/routes_styles.py            11      3    73%
src/maps4u/api/routes_system.py            65      6    91%
src/maps4u/api/routes_viability.py         29      3    90%
src/maps4u/bundle/__init__.py               0      0   100%
src/maps4u/bundle/archive.py              130     16    88%
src/maps4u/core/__init__.py                 0      0   100%
src/maps4u/core/config.py                  87      2    98%
src/maps4u/core/db.py                      64      1    98%
src/maps4u/core/errors.py                  26      0   100%
src/maps4u/core/ids.py                     27      3    89%
```

MAPS4U v1.0.3 - PYTEST EXECUTION & COVERAGE RECORD (PAGE 4 OF 4)

97 Automated Tests (Unit, Property, API, Golden Layout, v1.0.1 Defects, v1.0.2 Hardening, v1.0.3 Cartography) + Coverage

```
src/maps4u/core/logging.py               30      6    80%
src/maps4u/core/models.py                40      0   100%
src/maps4u/core/schemas.py               81      0   100%
src/maps4u/core/state.py                 95     13    86%
src/maps4u/geo/__init__.py                0      0   100%
src/maps4u/geo/acquire.py                34      0   100%
src/maps4u/geo/cache.py                 183     30    84%
src/maps4u/geo/corrections.py           120     16    87%
src/maps4u/geo/crs.py                    67     11    84%
src/maps4u/geo/features.py               49     11    78%
src/maps4u/geo/fetch.py                  89      5    94%
src/maps4u/geo/overpass.py               64      7    89%
src/maps4u/labels/__init__.py             0      0   100%
src/maps4u/labels/engine.py             241     24    90%
src/maps4u/layout/__init__.py             0      0   100%
src/maps4u/layout/furniture.py          164      1    99%
src/maps4u/layout/templates.py          131     11    92%
src/maps4u/qa/__init__.py                 0      0   100%
src/maps4u/qa/preflight.py               87      7    92%
src/maps4u/render/__init__.py             0      0   100%
src/maps4u/render/assess.py              85      3    96%
src/maps4u/render/cairo_canvas.py       210     13    94%
src/maps4u/render/canvas.py              59      7    88%
src/maps4u/render/proof_renderer.py      64      0   100%
src/maps4u/render/style_engine.py       235     28    88%
src/maps4u/worker/__init__.py             0      0   100%
src/maps4u/worker/celery_app.py           8      8     0%
src/maps4u/worker/executor.py           131      3    98%
src/maps4u/worker/pipeline.py            36      0   100%
src/maps4u/worker/tasks.py               40      0   100%
-----------------------------------------------------------
TOTAL                                  3238    291    91%
====================== 97 passed, 624 warnings in 44.88s =======================
```